

Transit Accessibility and Spatial Equity Analysis

Greater Portland, Maine | PostgreSQL/PostGIS, Python ETL, Spatial SQL, QGIS Automation
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TRANSIT STOPS	TRANSIT ROUTES	SIDEWALK SEGMENTS	NEIGHBORHOOD UNITS
982	416	10,670	7

Executive Summary

This project evaluates transit accessibility and spatial equity across a first-run Greater Portland analysis area. The pipeline downloads public GPCOG transit, route, sidewalk, and study-area layers, loads them into PostGIS, runs spatial SQL, and exports a QGIS map and report-ready findings.

The 800-meter transit stop buffer covers about 50,939 residents (69.12%) across the current neighborhood sample, leaving 22,761 residents outside that access band.

Important interpretation note: neighborhoods, schools, and hospitals are currently synthetic sample records. They keep the end-to-end workflow operational while verified public Census and facility datasets are added. Treat the workflow, SQL, and cartographic automation as production structure, and treat current neighborhood-level scores as demonstration values until those layers are replaced.

Strongest Accessibility Scores

Neighborhood	Population	Score	Transit	Sidewalk	School	Hospital	Nearest Stop m
Portland Peninsula	18,000	91.25	100	100	69.23	74.02	122.21
South Portland Waterfront	13,500	72.27	100	71.29	43.58	21.66	77.41
Westbrook Core	11,800	64.44	100	16.35	97.69	0	39.46

Lowest Accessibility Scores

Neighborhood	Population	Score	Transit	Sidewalk	School	Hospital	Nearest Stop m
Gorham Center	7,800	29.58	70	5.25	0	0	731.30
East Deering	7,200	31.51	70	11.70	0	0	416.32
Scarborough North	9,000	36.46	70	28.20	0	0	716.90

Map Output

The map was generated from PostGIS layers using the QGIS Python API. It includes transit routes, stops, sidewalks, schools, hospitals, study-area boundaries, graduated accessibility-score polygons, legend, scale bar, and north arrow.



Spatial Analysis Results

Transit Coverage Buffers

Buffer	Population Inside	Population Outside	Percent Covered
400 m	33,577	40,123	45.56%
800 m	50,939	22,761	69.12%

Accessibility Scores

Neighborhood	Population	Accessibility	Transit	Sidewalk	School	Hospital	Nearest Stop m
Portland Peninsula	18,000	91.25	100	100	69.23	74.02	122.21
South Portland Waterfront	13,500	72.27	100	71.29	43.58	21.66	77.41
Westbrook Core	11,800	64.44	100	16.35	97.69	0	39.46
Falmouth Foreside	6,400	52.13	100	27.58	19.27	0	169.19
Scarborough North	9,000	36.46	70	28.20	0	0	716.90
East Deering	7,200	31.51	70	11.70	0	0	416.32
Gorham Center	7,800	29.58	70	5.25	0	0	731.30

Underserved Area Ranking

Rank	Neighborhood	Score	Nearest Stop m	Score Below 40	Farther Than 800 m
1	Gorham Center	29.58	731.30	1	0
2	East Deering	31.51	416.32	1	0
3	Scarborough North	36.46	716.90	1	0

Methods

The database uses geometry columns in EPSG:4326 and transforms geometries to EPSG:26919 for meter-based analysis. Transit stop buffers are created with **ST_Buffer** and dissolved with **ST_Union**. Population coverage is estimated by intersecting neighborhood polygons with 400-meter and 800-meter buffers. Nearest stop and facility distances are calculated from neighborhood centroids with projected **ST_Distance** measures and KNN-style ordering.

The accessibility score is a weighted index from 0 to 100: 40 percent transit access, 30 percent sidewalk access, 20 percent school access, and 10 percent hospital access. Underserved neighborhoods are flagged when the score is below 40, the nearest stop is more than 800 meters away, or the polygon is outside the 800-meter transit buffer.

Technical Summary

Layer	Records
Hospitals	3
Neighborhoods	7
Schools	5
Sidewalk segments	10,670
Study-area polygons	17
Transit routes	416
Transit stops	982

PostGIS version: POSTGIS="3.6.2 3.6.2" [EXTENSION] PGSQL="170" GEOS="3.14.1dev-CAPI-1.20.4" PROJ="8.2.1 NETWORK_ENABLED=OFF URL_ENDPOINT=USER_WRITABLE_DIRECTORY=C:\WINDOWS\ServiceProfiles\NetworkService\AppData\Local\proj" (compiled against PROJ 8.2.1) LIBXML="2.12.5" LIBJSON="0.12" LIBPROTOBUF="1.2.1" WAGYU="0.5.0 (Internal)" TOPOLOGY

Key SQL artifacts: coverage_buffers, coverage_summary, nearest_transit_stop, neighborhood_sidewalk_access, accessibility_scores, underserved_areas, and neighborhood_accessibility_map.

Recommendations

- Replace synthetic neighborhoods with Census block groups and ACS demographic attributes.
- Add verified public school and healthcare-facility sources before presenting equity findings as authoritative.

- Use the underserved ranking to prioritize candidate corridors for additional stop placement or service-frequency review.
- Add demographic overlays, including income, age, disability, and vehicle access, to deepen the equity interpretation.

Transit Accessibility and Spatial Equity Analysis for Greater Portland, Maine. Built with Python, PostgreSQL/PostGIS, spatial SQL, and QGIS automation.